

UNIVERSITY OF SCIENCE - VNUHCM
FACULTY OF INFORMATION TECHNOLOGY



OBJECT-ORIENTED PROGRAMMING

AI - USAGELOG

SUBJECT: MANAGEMENT SYSTEM APPLICATION

Instructor:

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Students:

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Name	Task AI	Tool	Instruction	Student Verification or Modification
Duong Anh Hào	Build system and module dependency	Gemini	Fix CMakeLists.txt LNK2019 errors to properly link domain, repository, and service modules.	Updated the CMake configuration based on the suggestions, linked all project layers correctly, and successfully compiled the core modules without linker errors.
	Cross-Platform Compiler Standardization	Gemini	Configure CMake to force MSVC compilation with the C++20 standard and resolve compiler compatibility issues.	Applied the required MSVC compiler flag in CMake and removed hardcoded Qt paths to ensure the project builds correctly across different environments.
	Architecture Review & Data Encapsulation	Gemini	Evaluate the layered architecture and determine how to handle QString conversions without violating domain-layer separation.	Reviewed and modified the code to keep Qt-specific conversions inside implementation files, preventing Qt dependencies from leaking into domain entities.
	GUI Design & Code Documentation	Gemini	Review AddPackageDialog.cpp and EditPackageDialog.cpp, optimize QTabWidget initialization logic, and generate standard English Doxygen comments.	Applied only the safe refactoring suggestions and documentation improvements while preserving the existing domain data binding and GUI behavior.
	Debugging & Legacy Code Cleanup	Gemini	Fix application crash caused by filtering after removing the filter panel UI.	Identified and removed obsolete filter-related logic, simplified the filtering function, and confirmed the segmentation fault was fully resolved.
	IDE Toolchain Troubleshoot	Gemini	Resolve Visual Studio MSVC build issues caused by incorrect MinGW/Qt toolchain configuration.	Cleared the corrupted build cache, migrated the workflow to Qt Creator, selected the correct MSVC2022 64-bit Kit, and verified

				a stable build environment.
	Data Model Refactoring & Query Update	Gemini	Asked where the repositories needed to be updated after replacing legacy boolean flags with exact Date objects in PackageQueryCriteria.	Used global search to trace obsolete references and manually updated SQL queries and in-memory filtering logic to handle importDate and exportDate.
	Memory Safety (Crash Fix)	Gemini	Asked to diagnose a Qt invalid nullptr parameter warning after removing the legacy FilterPanel.	Audited MainWindow.cpp, traced the event flow, and manually removed obsolete connections to ensure memory safety and prevent crashes.
Huỳnh Phúc Nguyễn	Report Outline and Heading Generation	Gemini	Generated a recommended report outline and English section headings after analyzing the project requirements and assessment guidelines.	Evaluated the suggested outline, removed overlapping sections, modified the heading hierarchy, and finalized a report structure appropriate for the team's Warehouse Management System project.
	Code Review & Codebase Optimization	Claude	Analyze the current Warehouse Management System codebase and identify areas for improvement, including redundant code, architecture issues, performance concerns, and possible refactoring opportunities.	Reviewed the AI suggestions against the actual source code, confirmed the identified issues manually, and selected only relevant improvements that aligned with the current architecture and project scope.
	Repository Architecture Review	Claude	Evaluate the coexistence of PackageFilter and PackageQueryCriteria-based filtering systems, and determine whether any components are obsolete or can be removed.	Checked the usage of PackageFilter throughout the codebase, confirmed that repository-level filtering through PackageQueryCriteria had replaced most previous functionality, and considered deletion only after ensuring no GUI dependency existed.(This has been changed considerably and now it has been deleted)
	Code	Claude	Identify duplicated logic	Compared both repository

	Duplication Reduction		between JsonPackageRepository and SqlitePackageRepository and suggest a cleaner architecture following OOP principles.	implementations manually, confirmed duplicated enum/date conversion helpers, and evaluated extracting shared helper functions without changing existing repository behavior.
Lâm Hồng Hải Hoàng Lê	GUI Design	Gemini	A small snippet of C++ code that uses Qt to create a pie chart, and hovering over each pie slice makes the relevant data pop up. Each slice should be able to store dynamic data	Using the template provided, modified and integrated into the program's GUI
	Debugging	Gemini	Tooltip flickers when hovered and disappears after a few seconds. Provided faulty code	Applied the fixes. The fixes do not affect other parts of the code
	GUI Design	Gemini	Make the progress bar vertical and centered, with popup hover, and without making a new class. Provided existing progress bar code	Modified the code to match with the existing GUI design, tweaked more settings
	GUI Design	Gemini	Change table's row background when highlighted, and disable border highlighting specific cells	Gemini provided several options, applied one of the fixes. The fixes do not affect other parts of the code
Nguyễn Viết Bách	Design Repository Interface	Antigravity	Design an abstract C++ repository interface IPackageRepository with CRUD and batch query methods for a Package entity	Reviewed method signatures and adjusted return types; Added query overloads for filtering by state and date range; Ensured interface is implementation-agnostic (no Qt/SQL leakage).

	JSON Repository Implementation	Antigravity	Implement IPackageRepository storing packages as JSON using nlohmann/json. Support add, update, remove, findById, findAll.	Fixed round-trip for packages with special characters in description; Added error handling for missing or corrupted JSON files.
	Table Model Implementation	Antigravity	Implement QAbstractTableModel for Package with 9 columns: ID, Name, Description, Category, Weight, Zone, Status, Import Date, Export Date. Add a 4-column compact variant.	Added ForegroundRole color coding per package status; Verified data() output matches domain Package fields exactly; Fixed date formatting to use zero-padded YYYY-MM-DD strings
	Main Window Build-up	Antigravity	Build Qt6 MainWindow with sidebar navigation (Dashboard, Inventory, Operations, Reports) and QStackedWidget page switching. Wire all slots to WarehouseGateway signals.	Customized layout, color palette, and button styles; Connected all CRUD and state-transition slots. Fixed slot wiring for selection-changed to enable/disable action buttons.
	Edit Dialog Creation	Antigravity	Create EditPackageDialog in Qt6 pre-filled with existing Package fields. Return updated Package data on accept.	Verified pre-fill logic for all field types including QDateTimeEdit for date fields. Fixed category ComboBox mapping to correctly match domain Category enum values.
	Filtering files removal	Antigravity	Delete files only related to filter that are no longer needed.	Confirmed the files were no longer referenced before deleting FilterPanel, FilterableHeaderView, MultiColumnFilterProxy, PackageFilterDialog, and PackageFilter.

	CMake reconfiguration	Antigravity	Project does not build	Identified stale CMake autogen cache, reconfigured the project, regenerated build files, and verified a successful build.
Đỗ Minh Khang	Code Review and Software Architecture	Claude	Reviewed the assignment specification and existing backend codebase to identify implemented design patterns, evaluate OOP compliance, and determine missing patterns required for the GUI layer.	Cross-checked every identified pattern and OOP requirement against the actual source code and assignment specification before using the results to guide later development.
	Back-end development	Claude	Designed and implemented a SQLite-backed repository alongside the existing JSON repository, including schema, connection handling, CRUD operations, criteria-based filtering, and later support for an additional database field.	Built and tested the implementation using a real SQLite database, verified CRUD operations and schema consistency, and confirmed the new field integrated correctly.
	Debugging	Claude	Performed a full codebase audit to identify compile-time errors, runtime issues, and logical bugs, then suggested fixes for each issue.	Applied and tested the fixes locally, confirming successful compilation and ensuring no regressions were introduced.
	Build Configuration and Debugging	Claude	Generated portable CMake configuration files and diagnosed build/runtime issues including Qt configuration, missing DLLs, SQL driver loading, and database initialization problems.	Applied each fix incrementally and confirmed the project built and executed successfully after every change.
	Code Review	Claude	Reviewed a teammate's repository-related feature branch to evaluate implementation correctness and determine whether it was safe to merge.	Confirmed the reported issue by inspecting the codebase, communicated the required fix, and verified the implementation before merge.

	Backend Design	Claude	Redesigned the query interface to restore flexible multi-condition querying after migration to a criteria-based SQL repository.	Implemented the suggested changes and tested multiple query combinations to ensure behavior matched the previous implementation.
	GUI design and software architecture	Claude	Reviewed the GUI implementation for proper Observer pattern usage, then generated the required Observer-based architecture and integrated it into the existing GUI.	Integrated the generated code, resolved build issues, and confirmed the updated GUI functioned correctly without affecting existing features.
	Documents	Claude	Generated UML class diagrams for the GUI layer based on the existing source code, including relationships between GUI components and the backend.	Compared the generated diagrams against the actual source files, corrected inconsistencies, and integrated the finalized diagrams into the project documentation.